

MULTIDISCIPLINARY DESIGN PROJECT
MOTION CONTROL ENGINEERING REPORT

MotionController Dynamic Braking Characterisation and Validation

Experimental basis, control abstraction, and straight-motion validation of speed-dependent rear-wheel braking

Report date	9 September 2026
System	4-wheel robot; independently driven rear wheels with Hall encoders; servo-steered front wheels
Scope	AT8236 BRAKE/COAST actuation, brake-response characterisation, DynamicBrakeMap, WheelSpeedController integration, and straight-motion validation
Primary brake-map experiment	8 Sep 2026, 14:27 SGT; battery 10.95 V; rear wheels lifted
Straight-motion validation	8 Sep 2026: 2000-CPS repeat batch at 10.95 V; 9 Sep 2026: cross-speed sweep, battery 11.95 V at 11:48 SGT
Engineering decision	Retain proportional dynamic braking for straight motion and proceed to curved-motion implementation

Purpose. Document the engineering reasoning and experimental evidence behind the proportional dynamic-braking feature added to the rear-wheel control stack. The report records why a speed-dependent mapping was required, how braking was integrated without collapsing software layers, and whether the resulting controller remained well behaved across the straight-motion operating range tested on 8–9 September 2026.

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1 Executive Summary

The AT8236 rear-wheel H-bridge supports a BRAKE state when both control inputs are high and a coast/high-impedance state when both are low. The low-level motor driver therefore implements proportional dynamic braking by applying identical in-phase PWM to both bridge inputs: the high portion of each PWM cycle commands BRAKE and the low portion commands COAST. The resulting physical duty is a brake-state duty cycle, not a percentage of braking torque.

Initial experiments showed that useful braking is strongly speed dependent. At approximately 2000 CPS, duty values below roughly 85–90% had little effect, while 100% braking sharply reduced wheel run-on. At lower wheel speeds, even higher physical duty was required to obtain the same relative reduction in run-on. A single fixed conversion from logical brake request to PWM was therefore rejected.

A separate `DynamicBrakeMap` abstraction was introduced between `WheelSpeedController` and `rwdriver`. `WheelSpeedController` produces a normalized `brakeDemand` in the range 0–1; `DynamicBrakeMap` converts `brakeDemand` and measured wheel speed into physical brake PWM using an experimentally derived two-dimensional table. `WheelSpeedController` engages proportional braking only when the wheel is moving in the requested direction and exceeds its target speed. The initial overspeed thresholds are 200 CPS to engage, 100 CPS to release, and 800 CPS for full normalized demand.

At 2000 CPS, three 1000 mm validation runs at 10.95 V ended at +1.237, +1.699, and +0.842 mm. The mean endpoint error was +1.259 mm with sample standard deviation 0.429 mm. This was descriptively tighter than the immediately preceding selective-deadband-only batch at the same battery voltage, which had mean +1.897 mm and sample standard deviation 1.605 mm. The comparison is based on only three runs per condition and is not treated as a formal statistical-significance result.

A subsequent 9 Sep speed sweep, with battery measured at 11.95 V at 11:48 SGT, produced final errors of +1.231 mm at 1000 CPS, +1.501 mm at 3000 CPS, +0.578 mm at 4000 CPS, and +1.303/+1.765 mm for two nominal 5000-CPS runs. The 5000-CPS commands generated triangular profiles and reached only about 4.45 kCPS. No instability or excessive braking was observed above the 2000-CPS upper calibration speed, where the current map deliberately clamps to its 2000-CPS row.

Key findings

Finding	Evidence	Interpretation
Brake effect is speed dependent	The physical PWM needed for the same normalized effect falls as speed rises; e.g. 0.50 demand maps to about 97.4% at 500 CPS but 89.8% at 2000 CPS.	A fixed brake-demand-to-PWM law is not appropriate.
Layered abstraction is useful	<code>WheelSpeedController</code> produces normalized demand; <code>DynamicBrakeMap</code> performs empirical mapping; <code>rwdriver</code> exposes physical BRAKE/COAST duty.	Control intent remains separated from hardware actuation.
2000-CPS endpoint spread reduced	Dynamic-brake batch: mean +1.259 mm, SD 0.429 mm; previous selective-deadband batch: mean +1.897 mm, SD 1.605 mm.	Active braking appears to suppress poor deceleration tails, although $n = 3$ is too small for a formal significance claim.
High-speed behaviour remained stable	1000–5000 CPS nominal sweep produced +0.58 to +1.76 mm endpoint errors; 5000-CPS cases peaked near 4.45 kCPS.	Clamping the physical map above 2000 CPS was acceptable in the tested straight-motion cases.
Braking remains per-wheel	Logs show one wheel can enter BRAKE while the other remains in DRIVE.	The controller can correct overspeed locally without imposing artificial symmetric braking.

Table 1: Summary of engineering findings.

Decision: the dynamic-braking architecture is sufficiently validated for straight motion. No additional straight-line brake tuning is required before beginning curved-motion implementation.

2 System Context and Control Objective

The robot uses two independently driven rear wheels and servo-steered front wheels. Each rear motor is controlled by its own `WheelSpeedController`, while `MotionController` supplies motion-profile targets, synchronisation corrections, and steering/heading commands. The addition of a deceleration profile reduced endpoint overshoot substantially, but the remaining stopping error still depended on how much wheel speed persisted near the endpoint.

The objective of this work was not to replace the existing feedforward + feedback wheel-speed controller. It was to add a distinct actuator mode for cases where reducing propulsion alone was insufficient: when a wheel was still moving in the requested direction but faster than its current profile target.

The required actuator semantics were therefore:

- positive/negative PWM for propulsion;
- explicit coast for no active torque;
- proportional dynamic braking for controlled deceleration; and
- full dynamic braking for target zero and final stopping.

3 H-Bridge Braking Semantics

For the AT8236 topology used on the controller board, the relevant logic states are BRAKE when both H-bridge inputs are high and COAST when both are low. This allows a proportional BRAKE/COAST waveform without commanding reverse propulsion.

```
PWM high:  IN1 = 1, IN2 = 1 -> BRAKE
PWM low:   IN1 = 0, IN2 = 0 -> COAST
```

`rwdriver` therefore exposes `DCMotor_SetBrakePWM(motor, brakePercent)`. Both H-bridge control inputs receive the same in-phase PWM. A `brakePercent` of 0 maps to coast, intermediate values alternate BRAKE and COAST within each PWM cycle, and 100% gives continuous dynamic braking. The percentage describes bridge-state duty only; it is not a calibrated torque or current percentage.

Reverse propulsion was deliberately rejected as a “gentle brake” because it would be plugging rather than dynamic braking and could produce unnecessary current and mechanical stress. The higher-level controller is also prevented from crossing through zero into reverse propulsion merely to correct ordinary overspeed while the wheel is already moving in the target direction.

4 Brake-Response Characterisation

4.1 Coarse and focused duty sweeps

A coarse automated test on 8 Sep 2026 at 13:16 SGT and 11.96 V applied 65% drive, allowed the wheel speed to settle, and then commanded brake duty from 0 to 100% in 10% steps. Measured run-on showed little convincing effect from 0–80%, an intermediate response near 90%, and a strong response at 100%.

A focused 80–100% sweep with 2.5% increments and repeated trials then showed a smooth monotonic reduction in run-on over the useful high-duty region. At roughly 2000 CPS, mean average-wheel run-on fell from about 502 counts at 80% braking to about 102 counts at 100%. This established that proportional braking was controllable, but it did not establish speed invariance.

4.2 Two-dimensional speed/duty experiment

The decisive characterisation was collected on 8 Sep 2026 at 14:27 SGT with battery voltage 10.95 V. Rear wheels were lifted. Five target speeds (500, 750, 1000, 1500, and 2000 CPS) were crossed with eight physical brake duties (0, 85, 87.5, 90, 92.5, 95, 97.5, and 100%), with three repetitions per combination. The experiment produced 120 trial summaries and 4080 trace samples.

The result was unambiguous: the same physical brake duty produces a much larger reduction in run-on at high speed than at low speed. For example, 95% brake duty reduced run-on by only about 9.8% relative to coast at 500 CPS, but by about 66.6% at 2000 CPS. Consequently, a single physical PWM threshold cannot represent a consistent logical braking command over the operating range.

Speed	Coast	85	87.5	90	92.5	95	97.5	100
500	35.8	36.0	35.3	34.7	32.8	32.3	25.5	15.8
750	86.2	83.2	82.3	80.2	75.3	63.8	47.5	29.7
1000	153.2	141.5	143.0	133.2	121.8	95.2	68.0	41.0
1500	332.0	300.8	285.5	242.3	191.8	143.7	101.7	67.5
2000	553.3	450.5	387.5	317.7	248.3	184.8	135.8	93.0

Table 2: Mean average-wheel run-on (encoder counts) in the 8 Sep 14:27 lifted-wheel brake-map experiment.

4.3 Normalised actuator effect and DynamicBrakeMap

To separate logical controller demand from physical bridge duty, the measured run-on was normalized between coast and full brake at each speed:

$$u_{\text{effect}}(B, v) = \frac{D_{\text{coast}}(v) - D_B(v)}{D_{\text{coast}}(v) - D_{100}(v)}. \quad (1)$$

This quantity is an empirical normalized reduction in lifted-wheel run-on. It is not instantaneous deceleration, braking torque, or current. Interpolation of the measured response yielded the physical PWM table used by DynamicBrakeMap.

Speed (CPS)	Demand 0.25	0.50	0.75	1.00
500	95.6	97.4	98.7	100.0
750	93.2	95.9	98.0	100.0
1000	91.8	94.8	97.4	100.0
1500	88.6	92.1	95.6	100.0
2000	85.5	89.8	94.1	100.0

Table 3: DynamicBrakeMap physical PWM calibration derived from normalized run-on reduction.

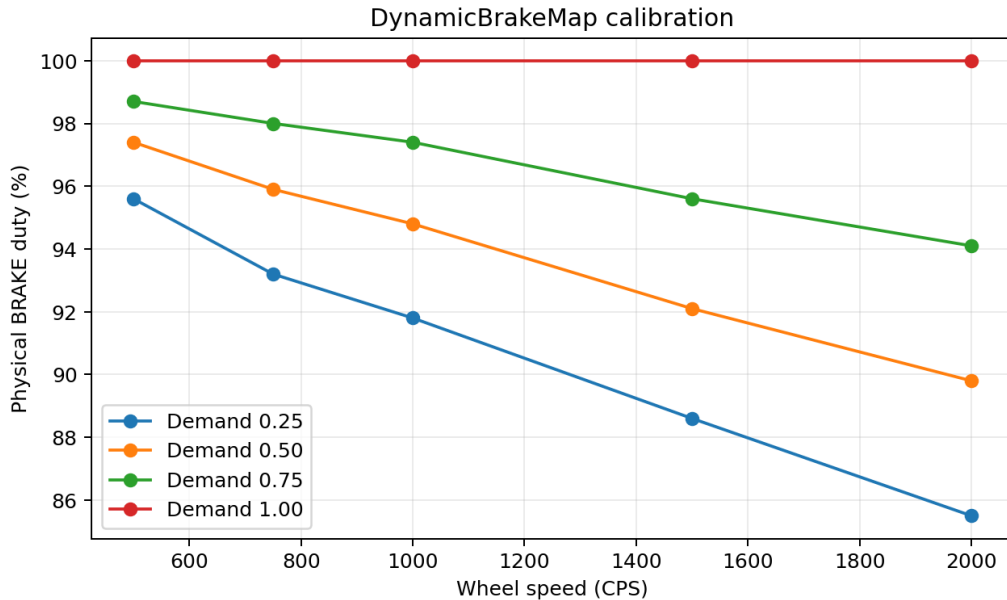


Figure 1: Physical BRAKE duty required by the calibrated map. Lower wheel speeds require higher duty for the same normalized demand.

Demand zero is special: it maps to true coast rather than to the first calibrated high-duty point. Any positive demand below the first calibrated knot (0.25) is promoted to 0.25. Speed is treated by

magnitude and clamped outside the 500–2000 CPS calibration range; the map does not extrapolate beyond measured speed knots.

5 Control Abstraction and Software Integration

5.1 Layering

The implementation deliberately keeps three responsibilities separate:

<code>WheelSpeedController</code>	→	<code>brakeDemand</code> [0..1]
<code>DynamicBrakeMap</code>	→	physical brake PWM [%]
<code>rwdriver</code>	→	BRAKE/COAST H-bridge waveform

`WheelSpeedController` therefore expresses control intent without knowing that useful physical braking occupies a narrow high-duty region. `DynamicBrakeMap` owns the empirical actuator calibration. `rwdriver` remains a hardware abstraction and accepts only physical percentages.

5.2 Overspeed-triggered braking law

Proportional braking is considered only when the wheel is moving in the commanded direction. The overspeed quantity is the excess speed magnitude:

$$e_b = |v_{\text{measured}}| - |v_{\text{target}}|. \quad (2)$$

The initial controller tuning uses a 200 CPS engage threshold, a 100 CPS release threshold, and 800 CPS overspeed for full normalized demand. Hysteresis is important because the empirical map has a deliberate discontinuity between demand zero (coast) and the first useful positive demand (0.25). Once braking is latched, the controller may remain at the minimum demand while overspeed lies between the release and engage thresholds.

On entry to proportional braking, the propulsion PID is reset and its update is skipped while the brake actuator is active. This avoids integrating a propulsion error while a different actuator mode is controlling the wheel. When braking releases, propulsion restarts from feedforward plus fresh feedback state.

Target zero remains a distinct full-stop condition: both wheels receive continuous full dynamic brake. `MotionController` subsequently returns to IDLE after the stationary condition is satisfied, at which point `WheelSpeedController_Stop()` retains its explicit coast semantics.

5.3 Telemetry and test-harness refinement

Per-wheel telemetry was extended with `actuatorMode`, `brakeDemand`, and `brakePWM` in addition to the existing propulsion PWM. This makes it possible to distinguish DRIVE, BRAKE, and COAST explicitly in debugger-exported traces.

For the 9 Sep speed sweep, storing the full telemetry structure every 10 ms would have exceeded the available RAM at low speed. Diagnostic storage was therefore decimated to 20 ms while the control loop and test-distance integration remained at 10 ms. This preserves control behaviour and final displayed distance while reducing the memory cost of the telemetry array. A one-cycle 10 ms brake event can therefore be missed by the stored trace, but not by the controller itself.

6 2000-CPS Straight-Motion Validation

Three 1000 mm runs were performed with dynamic braking enabled, nominal speed 2000 CPS, motion-profile acceleration 500 mm/s^2 , deceleration 250 mm/s^2 , heading $K_p = 0.80$, and wheel synchronisation $K_{sync} = 10$. Run 01 was recorded at 17:39 SGT on 8 Sep 2026 at 10.95 V; the battery was re-checked at 17:48 SGT and remained 10.95 V after the additional repeats. Exact run02/run03 acquisition times were not separately supplied.

Run	Final error (mm)	First BRAKING crossing (mm)	Post-BRAKING travel (mm)
Run 01	1.237	0.050	1.187
Run 02	1.699	0.314	1.385
Run 03	0.842	0.248	0.594
Mean	1.259	0.204	1.055
Sample SD	0.429		0.405

Table 4: 2000-CPS dynamic-braking straight-motion repeat batch.

The first run demonstrated the intended independent-wheel behaviour clearly. The right wheel first entered proportional braking about 12.8 mm before the endpoint, while the left wheel did not engage until about 2.85 mm before the endpoint. Brake demand remained low in normalized terms (roughly 0.25–0.36), while physical duty was high because the calibrated actuator map places useful dynamic braking in the 85–100% region.

Hysteresis behaved as designed: a wheel could enter BRAKE on an overspeed excursion, return to DRIVE after falling below the release threshold, and re-enter BRAKE if overspeed subsequently rose again. The two wheels did not need to brake simultaneously.

Compared with the immediately preceding selective-deadband-only batch at the same 10.95 V battery voltage, the new three-run batch had a lower mean endpoint error and a much smaller spread.

Batch	Battery	Mean final error (mm)	Sample SD (mm)	Interpretation
Selective deadband only (runs 07–09)	10.95 V	1.897	1.605	Reference im- me- di- ately be- fore ac- tive brak- ing
Dynamic braking (runs 01–03)	10.95 V	1.259	0.429	Lower mean and much tighter spread

Table 5: Descriptive comparison of endpoint statistics at 2000 CPS.

The sample size is too small to claim a formal statistical improvement, but the result is strong enough as engineering evidence to justify retaining the new brake path for broader validation.

7 Cross-Speed Validation on 9 Sep 2026

A subsequent sweep tested 1000 mm straight motions at nominal commands of 1000, 3000, 4000, and 5000 CPS, with a second 5000-CPS repeat. Battery voltage was measured at 11.95 V at 11:48 SGT. Individual exact acquisition times were not inferred from file metadata.

All tests retained the same profile acceleration and deceleration, heading gain, synchronisation gain, and brake-controller thresholds. The 2000-CPS three-run batch serves as the established reference point but was collected on the previous day at 10.95 V and therefore is not a matched voltage point within the 9 Sep sweep.

Nominal command	Peak profile (CPS)	Final error (mm)	First observed prop. brake (mm)	Max observed demand	Final yaw (deg)
1000	1000.0	1.231	995.427	0.364	−0.044
2000 ref.	2000.0	1.259	–	–	–
3000	3000.0	1.501	994.246	0.310	−0.281
4000	4000.0	0.578	Not observed	0.000	−0.049
5000 run 1	4450.8	1.303	991.740	0.292	0.035
5000 run 2	4454.5	1.765	989.366	0.458	−0.061

Table 6: Straight-motion results across the speed sweep. The 2000-CPS row is the mean of the previous-day reference batch.

Across the five newly uploaded 9 Sep runs, the mean final overshoot was approximately 1.28 mm, with a range from +0.58 to +1.76 mm. There was no obvious deterioration of longitudinal stopping accuracy with increasing commanded speed.

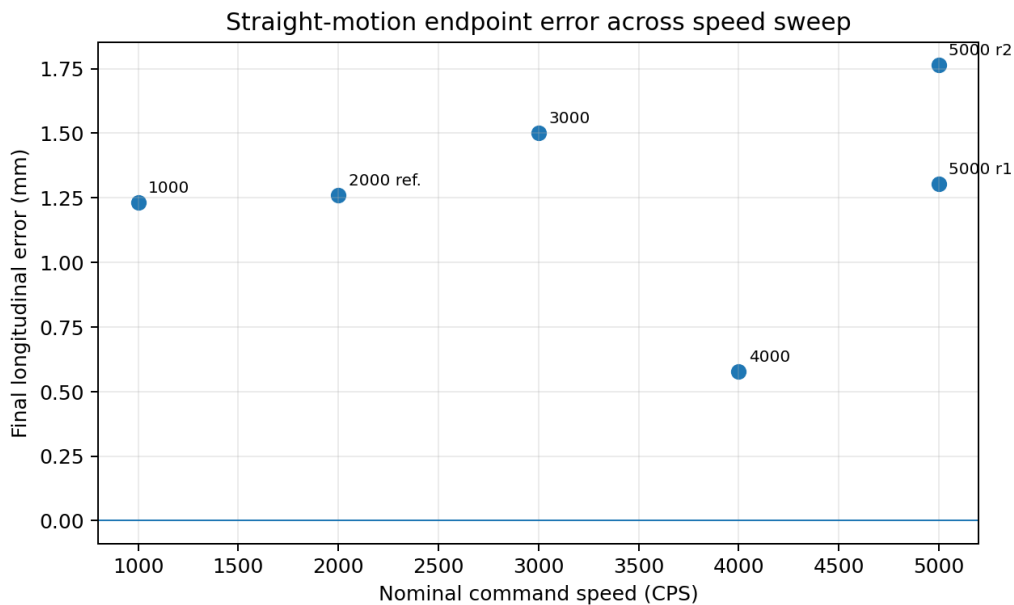


Figure 2: Final longitudinal error across the speed sweep. The 2000-CPS point is the previous-day three-run mean; the two 5000-CPS points are separate triangular-profile runs.

The nominal 5000-CPS tests require careful interpretation. With the fixed 1000 mm distance and current acceleration/deceleration limits, the profile is triangular: there is insufficient distance to reach 5000 CPS before deceleration must begin. The logged peak profile commands were approximately 4451 and 4455 CPS. These runs therefore provide evidence of stable high-speed profile execution, not evidence of steady 5000-CPS cruise tracking.

The 4000-CPS run is informative because no proportional-brake sample was observed before the final full-stop BRAKING state. This indicates that the new feature does not force unnecessary

braking when ordinary feedforward/feedback and the motion profile already produce an acceptable tail. Because telemetry was decimated to 20 ms, a single 10 ms proportional-brake event cannot be excluded.

The 3000-CPS run ended with about -0.281 degrees yaw, the largest final yaw in this small sweep. Longitudinal stopping remained accurate, but this illustrates that endpoint-distance validation and heading-quality validation are distinct objectives. Heading performance should continue to be evaluated separately as curved motion is introduced.

8 Engineering Interpretation

The main result is architectural as much as numerical. Dynamic braking is useful only in a narrow physical PWM region, and that region moves with wheel speed. Treating physical brake duty as if it were a linear controller output would therefore expose higher-level control code to hardware-specific nonlinearity. `DynamicBrakeMap` isolates that nonlinearity and gives `WheelSpeedController` a stable normalized actuator interface.

The cross-speed tests also support the decision to clamp, rather than extrapolate, above the 2000-CPS calibration range. At high wheel speed the existing 2000-CPS map row is already physically strong; extrapolating toward even lower physical duty would introduce an unmeasured relationship. Clamping is conservative in the sense of staying within characterised duty values, although the actual brake dynamics above 2000 CPS remain uncharacterised.

Independent per-wheel braking is compatible with the existing synchronisation and heading structure. The left and right wheel controllers may brake at different instants because their measured speeds and corrected targets differ. The observed logs show this asymmetry without producing an obvious longitudinal instability. This behaviour will be important for curved motion, where unequal wheel targets are intentional rather than merely corrective.

9 Limitations

1. The brake map is derived from lifted-wheel run-on. It is an actuator calibration proxy, not a direct model of chassis stopping distance, brake torque, motor current, or tyre-road force.
2. The physical map is characterised only from 500 to 2000 CPS. Behaviour above 2000 CPS has been validated functionally but not remapped experimentally.
3. The 8 Sep 2000-CPS repeat batch and the 9 Sep speed sweep were collected at different battery voltages. The 2000-CPS result is therefore a reference point, not a matched point within the 11.95 V speed sweep.
4. The validation sample sizes are small. Mean and standard-deviation comparisons are descriptive engineering evidence, not formal statistical equivalence or significance results.
5. The 20 ms diagnostic storage interval used on 9 Sep can miss a one-cycle 10 ms brake event even though the control loop itself still operates at 10 ms.
6. AT8236 current and thermal margins were not directly instrumented during these tests. Brake duty should not be interpreted as current percentage.
7. Curved motion will deliberately create unequal wheel targets and non-zero steering curvature. The present results establish straight-motion braking behaviour only.

10 Engineering Decision and Next Step

Decision: accept the current proportional dynamic-braking implementation for straight-motion development. Preserve the `DynamicBrakeMap` abstraction and the current 200/100/800 CPS overspeed tuning unless future data gives a specific reason to change it.

Development should now proceed to curved motion. The intended control layering is to generate non-zero left/right wheel relationships from steering curvature first, regulate wheel synchronisation around that kinematic relationship rather than around equal travel, and retain independent `WheelSpeedController` braking underneath. Heading correction must then be defined relative to the commanded curved path so that it does not fight the intended steering curvature.

A Experimental Record

Table 7: User-supplied acquisition/check times. File timestamps are not used as substitutes for actual collection times.

Date/time	Battery	Experiment	Record
8 Sep 13:16 SGT	11.96 V	Coarse 0–100% brake sweep	0–80% little effect; 90% intermediate; 100% strong
8 Sep, time not recorded	Not recorded	Focused 80–100% sweep	Established monotonic useful region; exact acquisition time/battery not supplied
8 Sep 14:27 SGT	10.95 V	2D brake map: 5 speeds $\times$ 8 duties $\times$ 3 reps	Primary calibration for DynamicBrakeMap
8 Sep 17:39 SGT	10.95 V	2000-CPS dynamic-braking run 01	First formal straight-motion validation run
8 Sep 17:48 SGT check	10.95 V	Battery re-check after additional repeat runs	No voltage change; exact run02/run03 acquisition times not separately supplied
9 Sep 11:48 SGT check	11.95 V	Cross-speed straight-motion sweep	Files: 1000, 3000, 4000, 5000 run01, 5000 run02; individual exact acquisition times not inferred

B Dataset Names Used for Final Speed Validation

- `mctrl_dyn_brake_1000.txt` – 1000 mm @ 1000 CPS; 447 stored samples at 20 ms.
- `mctrl_dyn_brake_run01/run02/run03` – 1000 mm @ 2000 CPS repeat batch; 10 ms telemetry.
- `mctrl_dyn_brake_3000.txt` – 1000 mm @ 3000 CPS; 195 stored samples at 20 ms.
- `mctrl_dyn_brake_4000.txt` – 1000 mm @ 4000 CPS; 181 stored samples at 20 ms.
- `mctrl_dyn_brake_5000_run01.txt` – nominal 5000 CPS triangular-profile test; 181 stored samples.
- `mctrl_dyn_brake_5000_run02.txt` – nominal 5000 CPS triangular-profile repeat; 177 stored samples.

C Configuration Snapshot

Parameter	Value
Motion distance	1000 mm
Motion-profile acceleration	500 mm/s ²
Motion-profile deceleration	250 mm/s ²
Heading controller K_p	0.80
Wheel synchronisation K_{sync}	10
WSC brake engage overspeed	200 CPS
WSC brake release overspeed	100 CPS
WSC full-demand overspeed	800 CPS
DynamicBrakeMap speed knots	500, 750, 1000, 1500, 2000 CPS
DynamicBrakeMap demand knots	0.25, 0.50, 0.75, 1.00
Control period	10 ms
9 Sep telemetry storage period	20 ms

Table 8: Controller and experiment configuration relevant to this report.